

***Generation Interconnection  
System Impact Study Report***

***For***

***PJM Generation Interconnection Request  
Queue Position AC1-087***

***“Siegfried-Hauto 69kV”  
3.8MW Capacity / 10MW Energy***

October 2017

## Preface

The intent of the Impact Study is to determine a plan, with approximate cost and construction time estimates, to connect the subject generation interconnection project to the PJM network at a location specified by the Interconnection Customer. As a requirement for interconnection, the Interconnection Customer may be responsible for the cost of constructing: Network Upgrades, which are facility additions, or upgrades to existing facilities, that are needed to maintain the reliability of the PJM system. All facilities required for interconnection of a generation interconnection project must be designed to meet the technical specifications (on PJM web site) for the appropriate transmission owner.

In some instances an Interconnection Customer may not be responsible for 100% of the identified network upgrade cost because other transmission network uses, e.g. another generation interconnection or merchant transmission upgrade, may also contribute to the need for the same network reinforcement.

The Study estimates do not include the Feasibility or Impact studies cost, or the time required to obtain property rights and permits for construction of the required facilities. The Interconnection Customer may be responsible for the right of way, real estate, and construction permit issues. For properties currently owned by transmission owners, the costs may be included in the study.

## General

PA Solar Park, LLC, the Interconnection Customer (IC), has proposed a Solar generating facility located in Carbon County of Pennsylvania, which is part of the PPL Electric Utilities (PPL EU) Central Region. The installed facilities will have a total capability of 10 MW, with 3.8 MW of this output being recognized by PJM as capacity. The proposed in-service date for this project, as requested by the IC during the project scoping meeting is June 2017.

AC1-087 is a re-submittal of the IC's prior queue project V3-040. Delays in permitting caused the V3-040 project to be withdrawn.

### **Point of Interconnection (POI)**

The IC requested a transmission level interconnection. As a result, AC1-087 will interconnect with the PPL EU transmission network via the Hauto-Siegfried #1 & #4 69kV lines. The AC1-087 POI is identical to that of V3-040. See Attachment 1 of this Study for more information.

### **Cost Summary**

The AC1-087 project will be responsible for the following costs:

| Description                                     | Total Cost       |
|-------------------------------------------------|------------------|
| Attachment Facilities                           | \$ 0             |
| Direct Connection Network Upgrades              | \$ 0             |
| Non Direct Connection Network Upgrades          | \$ 25,600        |
| Allocation for New System Upgrades              | \$ 0             |
| Contribution for Previously Identified Upgrades | \$ 0             |
| <b>Total Costs</b>                              | <b>\$ 25,600</b> |

## **Attachment Facilities**

There are no Attachment Facilities required as part of AC1-087. All Attachment Facilities were installed under V3-040.

## **Direct Connection Cost Estimate**

There are no Direct Connection costs required as part of AC1-087. All Facilities were installed under V3-040.

## **Non-Direct Connection Cost Estimate**

### Relay modifications at Hauto 69kV Substation

#### *PJM Network Upgrade Number n5904*

As part of prior Queue Project V3-040, PPL EU identified inconsistencies with the engineering package and relay settings. Revised drawings, description of relay and control, and additional information in response to PPL EU Relay Test questions are required. Therefore, the following activities need to be conducted to close out the coordination with PPL EU:

- Review the revised, most up to date documents from IC
- Review settings I/O configurations and ensure consistency with documents
- Revise settings at Hauto end as identified during V3-040
- Coordinate with PPL EU Relay Test to ensure there are no outstanding issues

The total preliminary cost estimate for the Non-Direct Connection work is \$25,600. These costs do not include CIAC Tax Gross-up.

The estimated schedule for the scope above is two (2) months from engineering start to construction finish, after the PJM three-party Interconnection Service Agreement (ISA) and Construction Service Agreement (CSA) are executed.

## **Interconnection Customer (IC) Requirements**

The IC requirements are identical to those specified under V3-040. Refer to the withdrawn queue V3-040 on the PJM website for more information.

## **Revenue Metering and SCADA Requirements**

### **PJM Requirements**

The IC will be required to install equipment necessary to provide Revenue Metering (KWH, KVARH) and real time data (KW, KVAR) for IC's generating Resource. See PJM Manuals M-01 and M-14D, and PJM Tariff Sections 24.1 and 24.2.

### **Meteorological Data Reporting Requirement**

The solar generation facility shall provide the Transmission Provider with site-specific meteorological data including:

- Temperature (degrees Fahrenheit)

- Atmospheric pressure (hectopascals)
- Irradiance
- Forced outage data

### **PPL Requirements**

The IC requirements are identical to those specified under V3-040. Refer to the withdrawn queue V3-040 on the PJM website for more information.

## **Transmission Network Impacts**

The Queue Project AC1-087 was evaluated as a 10.0 MW (Capacity 3.8 MW) injection into Siegfried 69kV substation in the PPL area. Project AC1-087 was evaluated for compliance with applicable reliability planning criteria (PJM, NERC, NERC Regional Reliability Councils, and Transmission Owners). Project AC1-087 was studied with a commercial probability of 100%. Potential network impacts were as follows:

### **Base Case Used**

Summer Peak Analysis – 2020 Case

### **Generator Deliverability**

*(Single or N-1 contingencies for the Capacity portion only of the interconnection)*

None

### **Multiple Facility Contingency**

*(Double Circuit Tower Line, Fault with a Stuck Breaker, and Bus Fault contingencies for the full energy output)*

None

### **Contribution to Previously Identified Overloads**

*(This project contributes to the following contingency overloads, i.e. "Network Impacts", identified for earlier generation or transmission interconnection projects in the PJM Queue)*

None

## **Summer Peak Load Flow Analysis Reinforcements**

### **New System Reinforcements**

*(Upgrades required to mitigate reliability criteria violations, i.e. Network Impacts, initially caused by the addition of this project generation)*

None

### **Contribution to Previously Identified System Reinforcements**

*(Overloads initially caused by prior Queue positions with additional contribution to overloading by this project. This project may have a % allocation cost responsibility which will be calculated and reported for the System Impact Study)*

*(Summary form of Cost allocation for transmission lines and transformers will be inserted here if any)*

None

### **Steady-State Voltage Requirements**

*(Results of the steady-state voltage studies should be inserted here)*

None.

**Short Circuit**

*(Summary of impacted circuit breakers)*

No issues identified.

**Stability and Reactive Power Requirement**

*(Results of the dynamic studies should be inserted here)*

No mitigations are required.

**Light Load Analysis - 2020**

Light Load Studies to be conducted during later study phases (as required by PJM Manual 14B).

Not Required.

**Facilities Study Estimate**

The estimated time to complete a Facilities Study is 7 months and the estimated cost is \$50,000.

**Delivery of Energy Portion of Interconnection Request**

*PJM also studied the delivery of the energy portion of this interconnection request. Any problems identified below are likely to result in operational restrictions to the project under study. The developer can proceed with network upgrades to eliminate the operational restriction at their discretion by submitting a Merchant Transmission Interconnection request.*

*Only the most severely overloaded conditions are listed. There is no guarantee of full delivery of energy for this project by fixing only the conditions listed in this section. With a Transmission Interconnection Request, a subsequent analysis will be performed, which will study all overload conditions associated with the overloaded element(s) identified.*

None.

Attachment 1 – Single Line Diagram

